

NSEJS 2026 FREE CRASH COURSE



BATCH DETAILS



Start: 14 September 2026



Grades: 8, 9 & 10



Sessions: 18 per subject



End: 31 December 2026



Timing: 5:30 AM – 6:30 AM



Batch: FREE

WATCH FULL VIDEO HERE!



▶ **SCAN & WATCH**
KNOW ALL DETAILS



WHAT'S INCLUDED?



High-Weightage Topics – Physics, Chemistry & Biology



8 Full-Syllabus Tests – Track your preparation



Faculty Feedback – Regular performance guidance



SAM Support – Continuous academic assistance



FILL THE FORM TO ENROLL YOURSELF
LINK IN DESCRIPTION



100% FREE BATCH



EXPERT FACULTY



BOOST YOUR PREPARATION



LIMITED SEATS!

Nidhi Prashar

Chemistry



- **16+ Years of teaching experience**
- **Gold Medallist in the Master's Degree (Chemistry), UGC NET Qualified**
- **IJSO EXPERT with 170+ NSEJS Selections**
- **Mentored Science Olympiads GOLD MEDALISTS**
- **Teaching and mentoring students IIT, NEET, NSEC, IJSO exams**
- **Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)**

REGISTRATION

Username

Email

Phone Number

Register



NSE-2026

REGISTRATION

HAS STARTED



APPLY NOW!

✓ NSEP

✓ NSEC

✓ NSEJS

✓ NSEB

✓ NSEA



REGISTRATION

Username

Email

Phone Number

Register



HOW TO REGISTER?

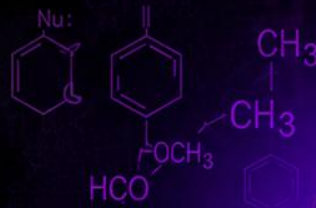
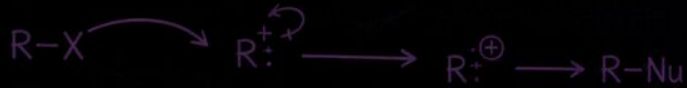
NSE 2026

— STEP BY STEP GUIDE —

NSEP • NSEC • NSEJS • NSEB • NSEA



30:04



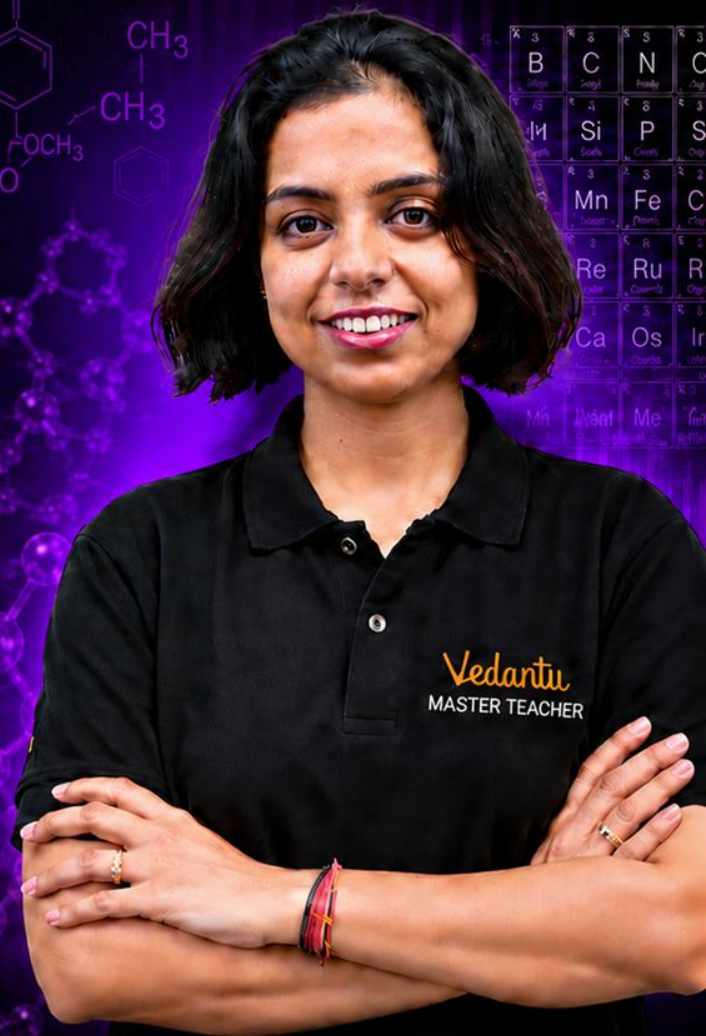
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30																											
B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																									
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Fr	Ra																										
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	Rb	Sr	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe																
Rf	Db	Sg	Lr	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr

NSEJS CHEMISTRY

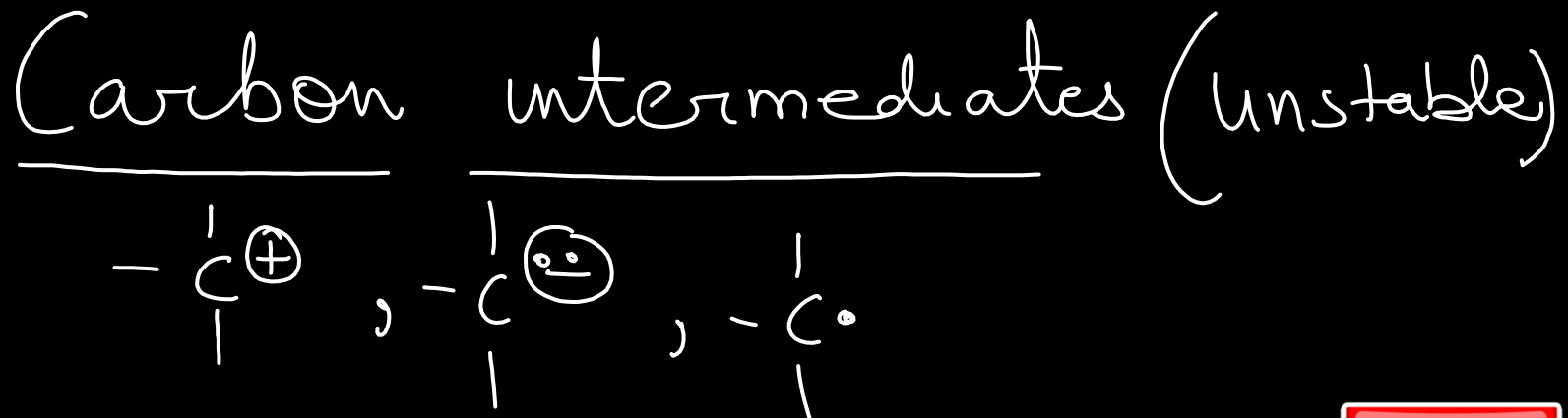
GOC:

REACTION INTERMEDIATES

ONE SHOT

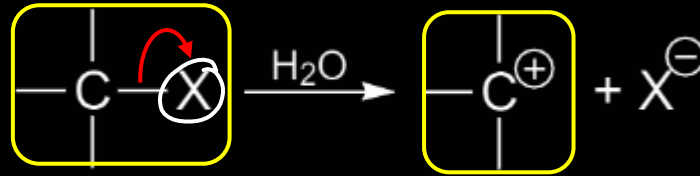


Reactive Intermediates

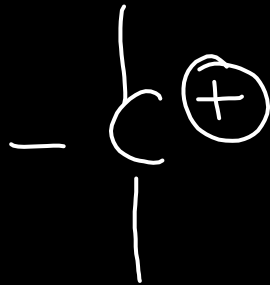


Reactive Intermediates

❖ Carbocation OR Carbonium Ion :



carbocation

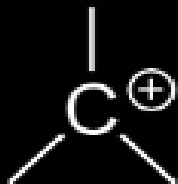


6e's (e^- deficient)

Electrophile

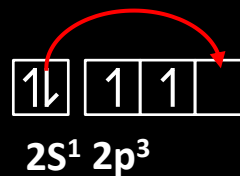
Reactive Intermediates

General Representation :



Electronic Configuration : ${}_6\text{C} - 1s^2 2s^2 2p^2$

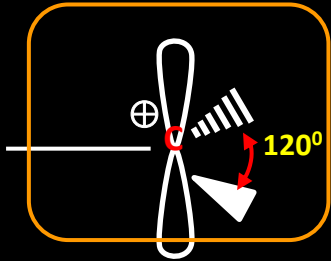
Ground state Configuration :



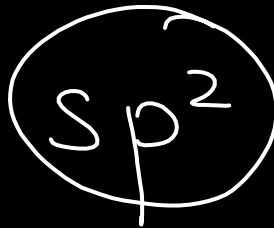
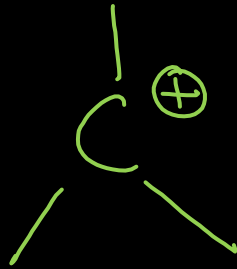
Reactive Intermediates

Excited state Configuration :

Geometry :



It is the short lived, highly reactive intermediate species in which C atom is carrying a positive charge, with 6e[⊖] s in it's valence shell.



Trigonal planar

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Reactive Intermediates :

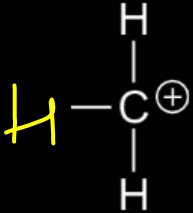
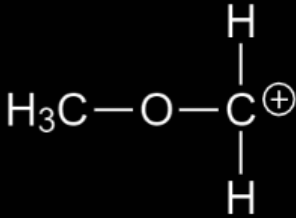
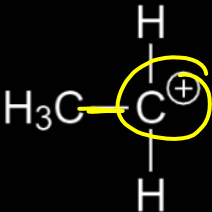
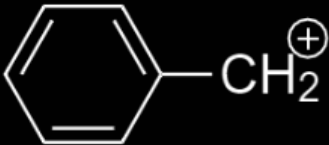
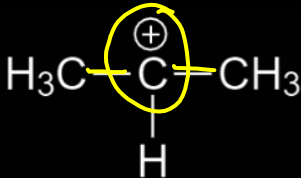
Characteristics :

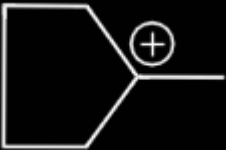
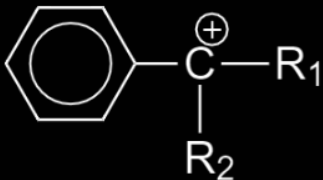
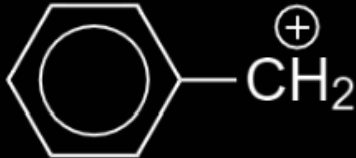
- 👉 The C is in sp^2 hybridized state with trigonal plane geometry.
- 👉 The cationic C has incomplete octet carrying $6e^-$ s in its valence shell.
- 👉 Since e^- deficient, behaves as a electrophile.
- 👉 Being electrophilic, accept e^- pair, thus a Lewis acid.

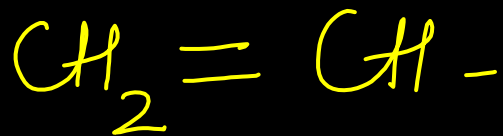
→ e^- pair acceptor

Types of Carbocation



Sr. No	Type	Example 1	Example 2
1.	Methyl $\text{C}^{\oplus}$		
2.	Primary $\text{C}^{\oplus}$ (1°)		
3.	Secondary $\text{C}^{\oplus}$ (2°)		

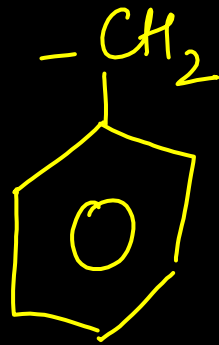
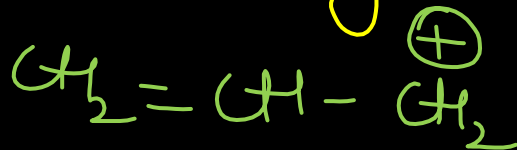
Sr. No	Type	Example 1	Example 2
4.	Tertiary $\text{C}^{\oplus}$ (3°)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}^{\oplus} \\ \\ \text{CH}_3 \end{array}$	
5.	Allylic $\text{C}^{\oplus}$	$\begin{array}{c} \text{R}_4 \quad \text{R}_3 \quad \text{R}_2 \quad \text{R}_1 \\ \diagup \quad \diagdown \quad \\ \text{C}=\text{C}-\text{C}^{\oplus} \end{array}$	$\text{H}_2\text{C}=\text{CH}-\text{CH}_2^{\oplus}$ OR 
6.	Benzylic $\text{C}^{\oplus}$		



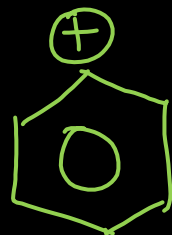
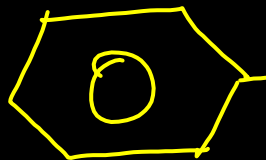
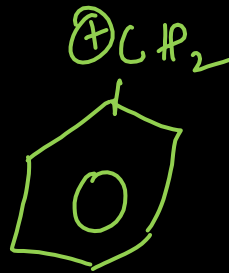
vinyl



allyl

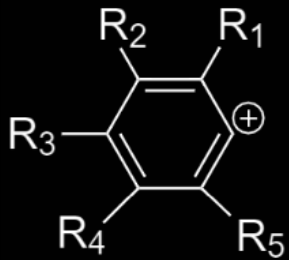


Benzyl



aryl

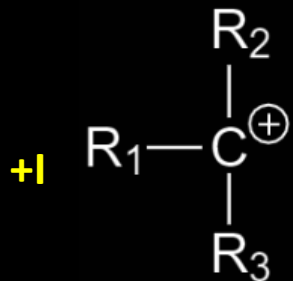
Types of Carbocation :

Sr. No	Type	Example 1	Example 2
7.	Vinylic $\text{C}^{\oplus}$	$\begin{array}{c} \text{R}_1 - \text{C} = \text{C}^{\oplus} \\ \quad \\ \text{R}_3 \quad \text{R}_1 \end{array}$	$\text{CH}_2 = \text{CH}^{\oplus}$
8.	Aryl $\text{C}^{\oplus}$		

Stability of Carbocation

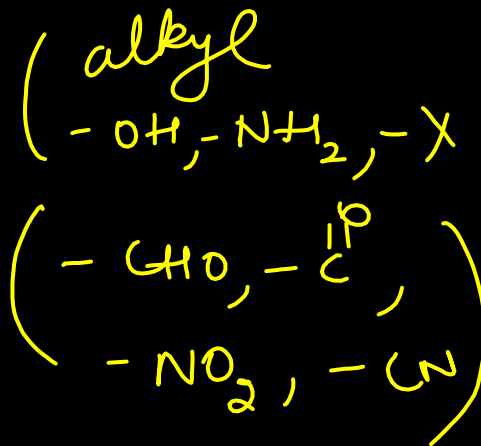
[Electronic effects contributing towards stability of carbocation]

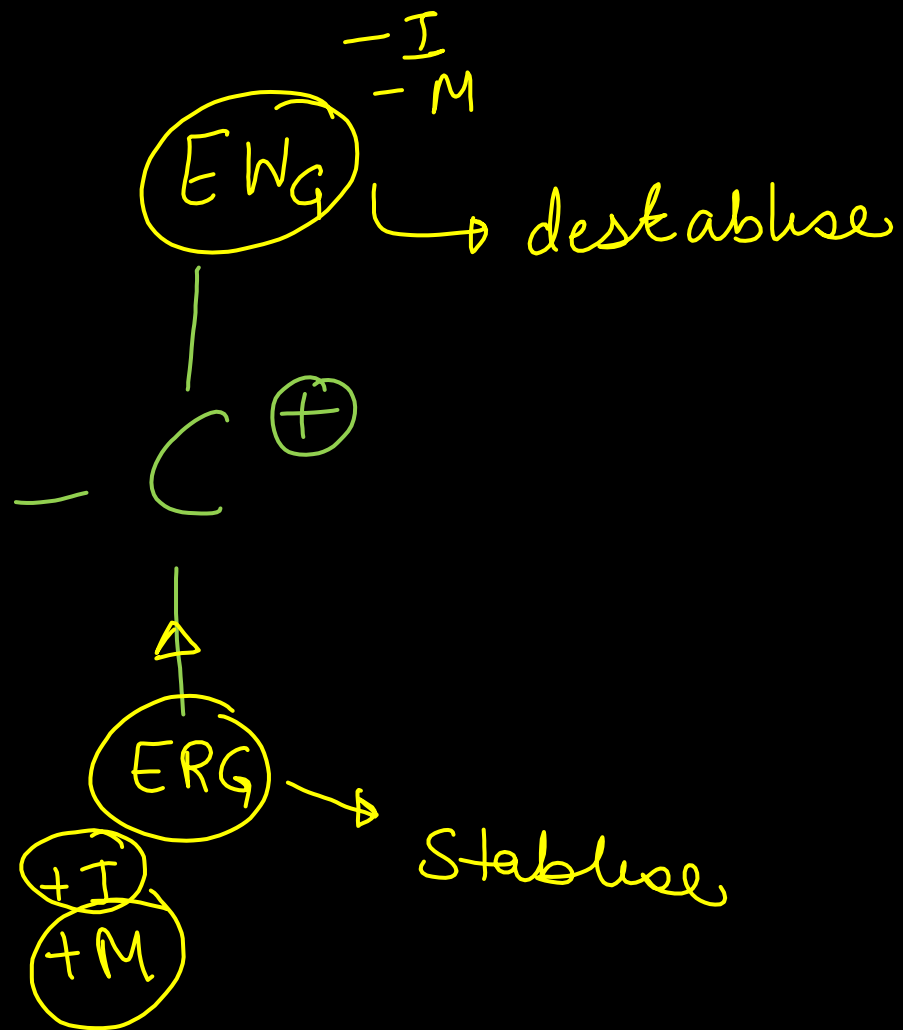
Inductive Effect :



EDG increases the stability of carbocation.

EWG decreases the stability of carbocation.

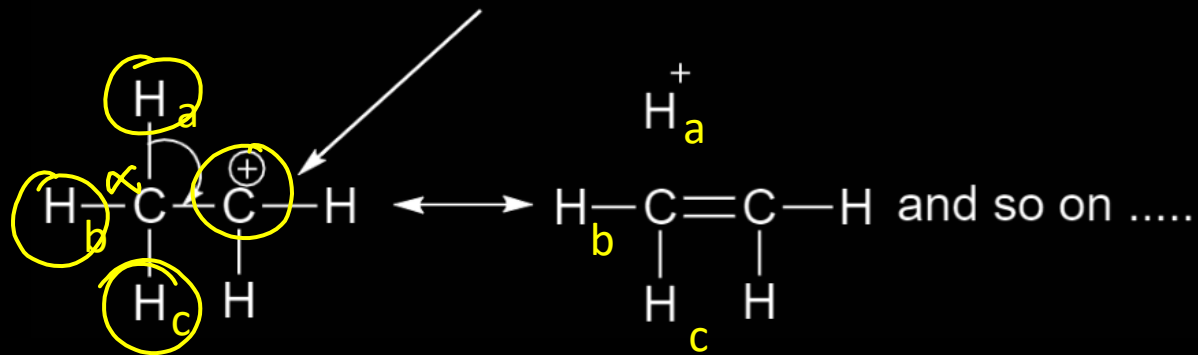




Stability of Carbocation

Hyperconjugation Effect :

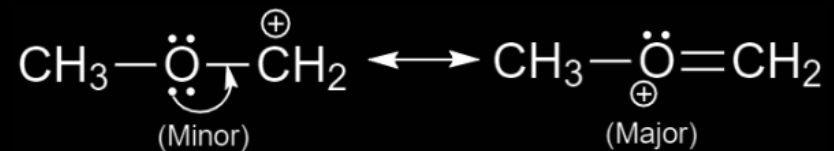
sp^2 hybridized C adjacent to sp^3 carbon.



“ Thus more the number of α -H's more is the stability”

Some Typical Carbocation

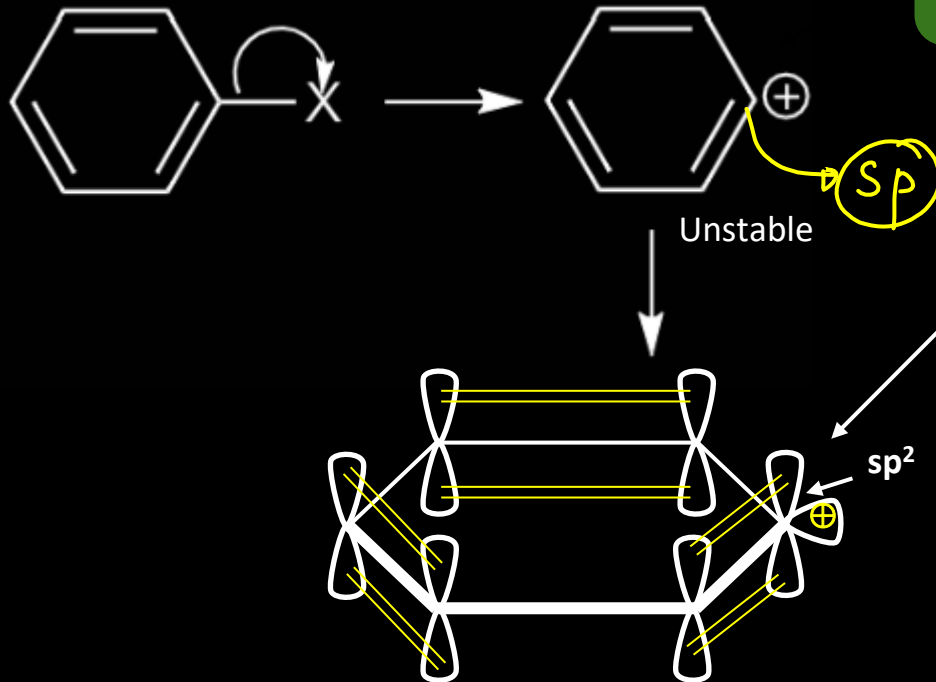
□ Resonance :



Some Typical Carbocation

Typical Cases :

Phenyl carbocation

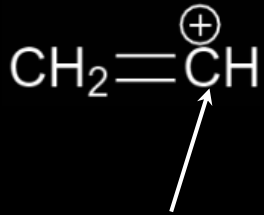


It would have been sp hybridized, should have been planar

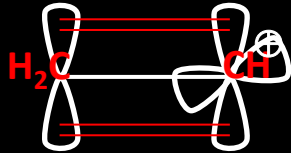
The positive charge is in vacant hybridized orbital which is perpendicular to the pure p-orbital forming π bond, hence can't delocalize to stabilize the cation.

Some Typical Carbocation

□ Vinyl Carbocation :



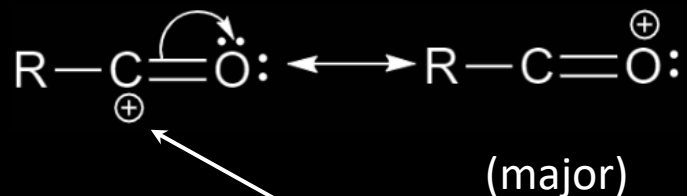
sp hybridised



The positive charge is not in conjugation with π bond also π bond and positive charge are in the orbitals which are perpendicular to each other.

Some Typical Carbocation

□ Acyl carbocation



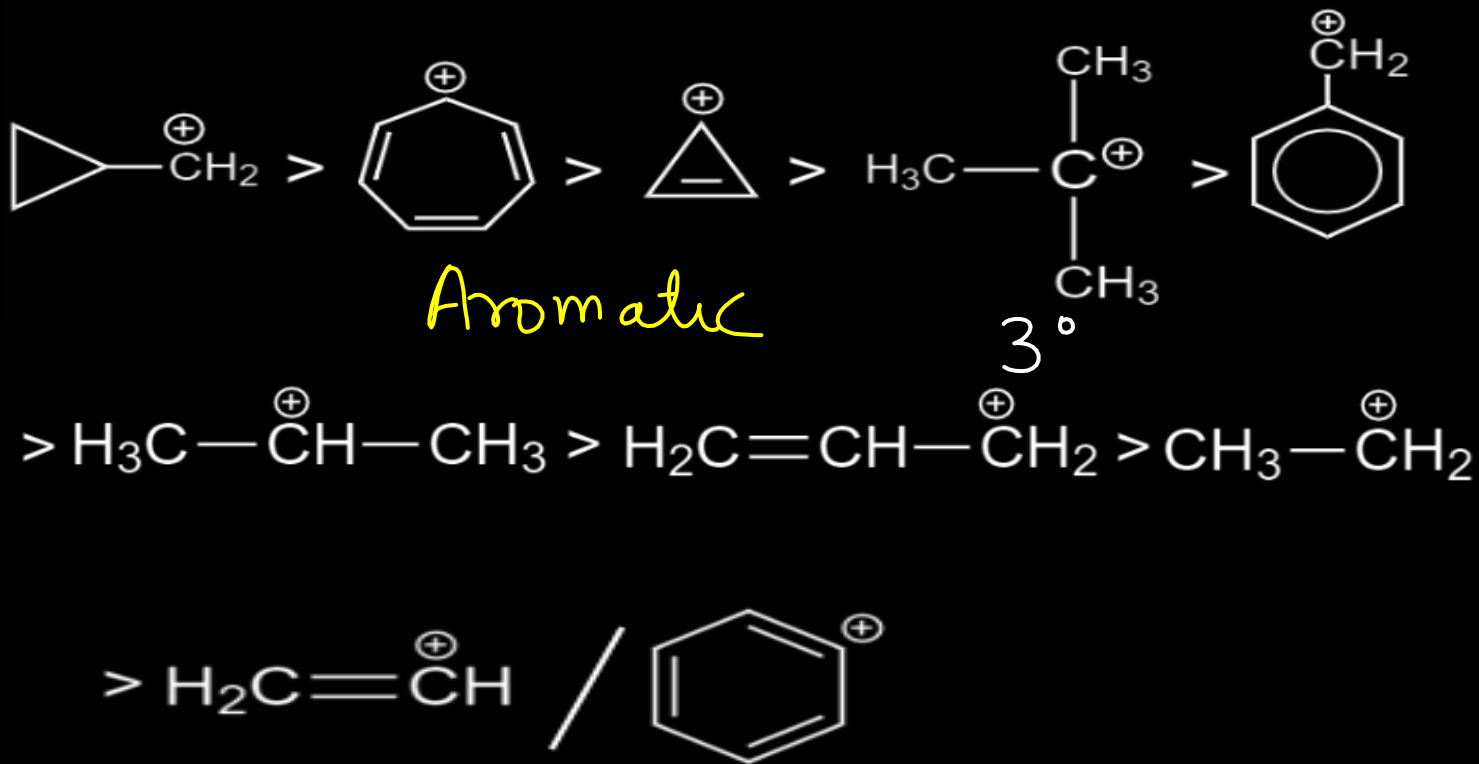
Acyl carbocation

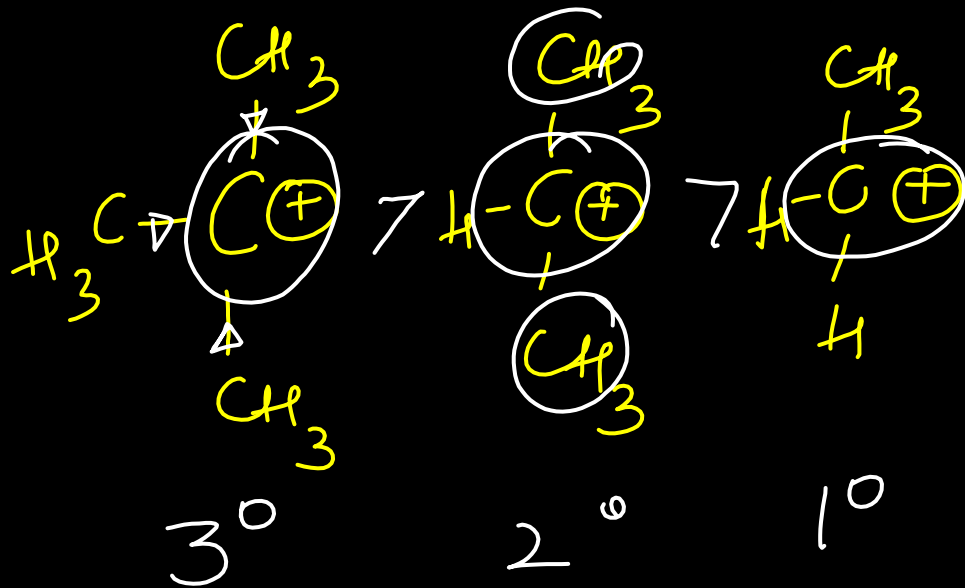
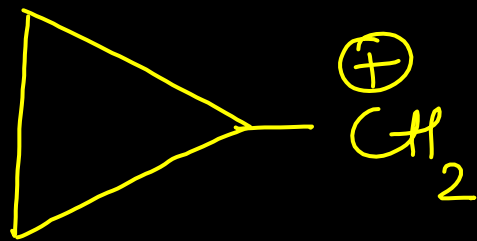
Stable

sp hybridized carbon
in conjugated with lone pair

Note : The stability of carbocation is measured by their 'heat of formation ΔH_f ' since it involves breaking of a bond, it is an endothermic process. further, more is The ΔH_f , lesser the stability of carbocation

General Order of stability of carbocation





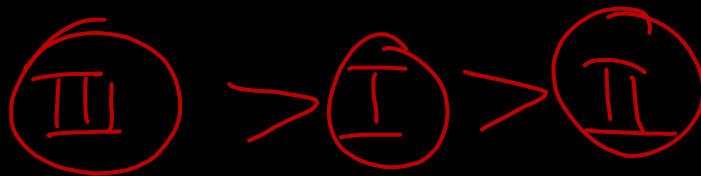
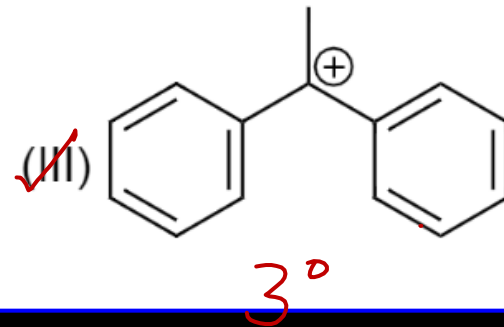
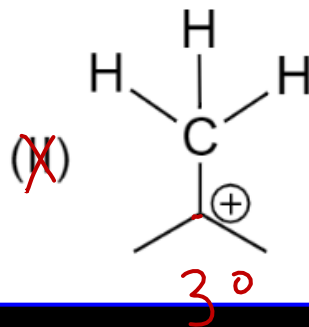
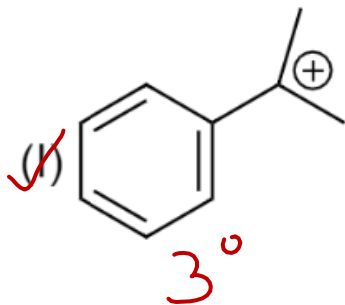
Very stable
DANCING Resonance

As per NCERT

$\Delta^{\oplus} > \text{Aromatic} > \text{Resonance}$

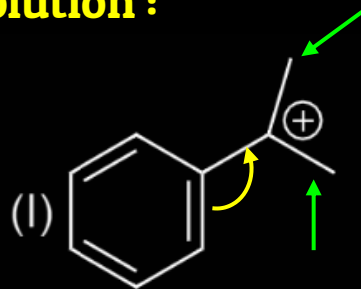
$3^{\circ} > 2^{\circ} > 1^{\circ} > \textcircled{+}\text{H}_3 > \text{vinyl} \approx$
 $\left\{ \begin{array}{l} \text{aryl} \\ \text{unstable} \end{array} \right.$

Write the order of stability of following carbocation

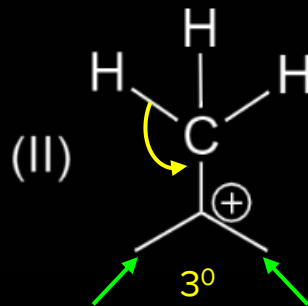


Write the order of stability of following carbocation

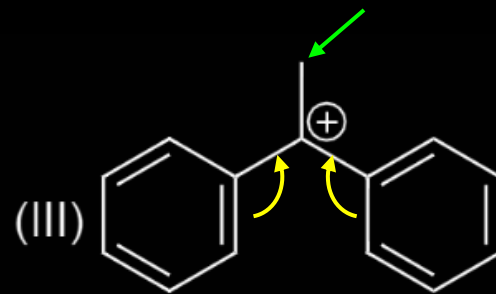
Solution :



3°
Resonance
6 α H
(Hyperconjugation)



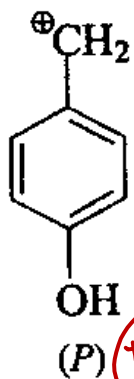
3°
9 α H
(Hyperconjugation)



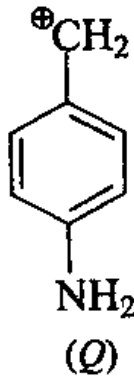
3°
Resonance
3 α H's
(Hyperconjugation)

The order is III > I > II

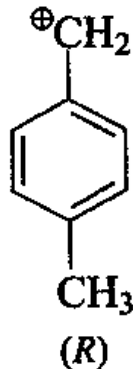
The decreasing order of stability of following cation is



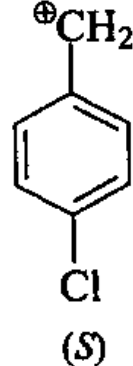
+M



+M



+I



A

P > Q > R > S

B

Q > S > R > P

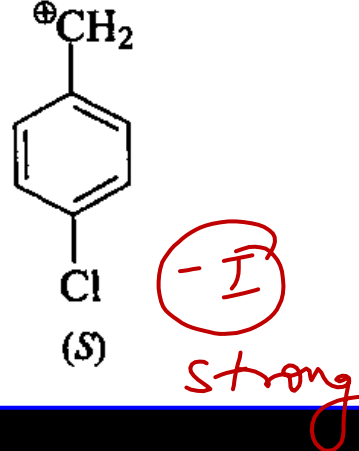
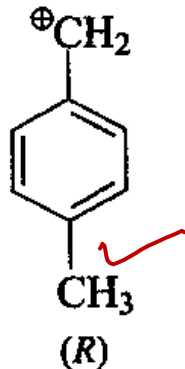
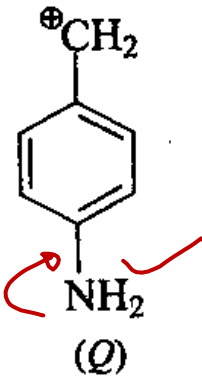
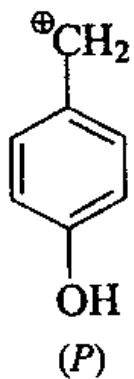
C

Q > P > S > R

D

Q > P > R > S

The decreasing order of stability of following cation is



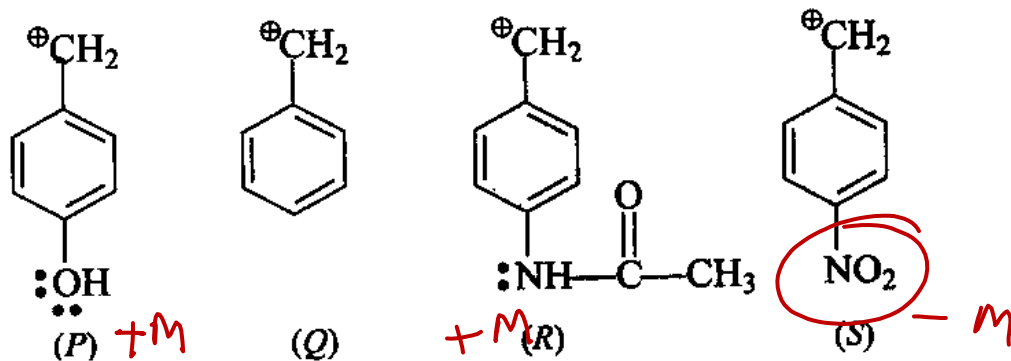
A P > Q > R > S

B Q > S > R > P

C Q > P > S > R

D Q > P > R > S

The decreasing order of stability of following cation is



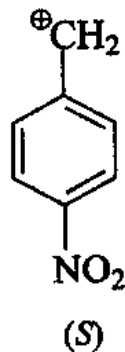
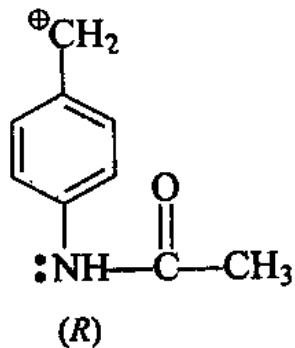
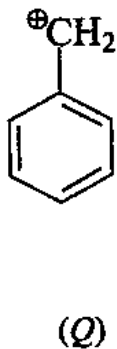
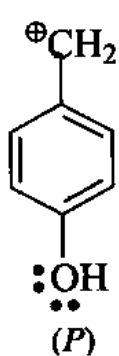
A $P > R > Q > S$

B $Q > R > S > P$

C $R > S > Q > P$

D $P > R > S > Q$

The decreasing order of stability of following cation is



A

P > R > Q > S

B

Q > R > S > P

C

R > S > Q > P

D

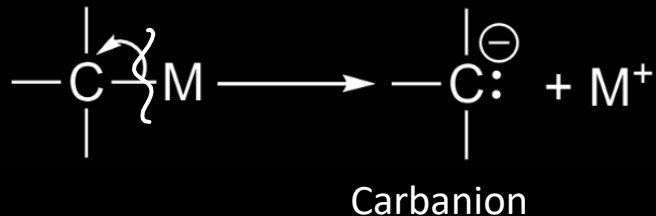
P > R > S > Q

Stability

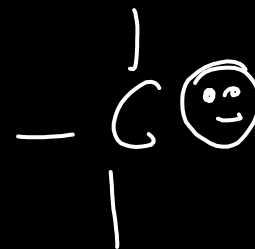
(FIR)

{ Inductive $+I/-I$
Resonance $(+M/-M)$
Hyperconj

Carbanion



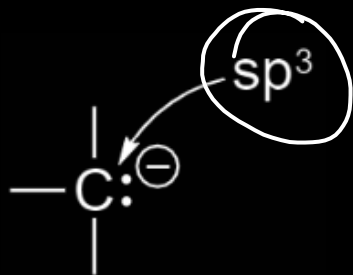
Where,
M = Metal



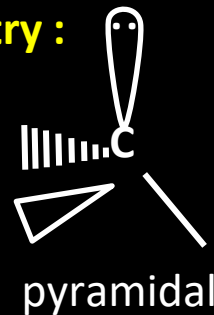
Definition

It is the short lived highly reactive intermediate species in which C atom is carrying negative charge with 8 e⁻s in the valence shell.

General Representation:



Geometry :



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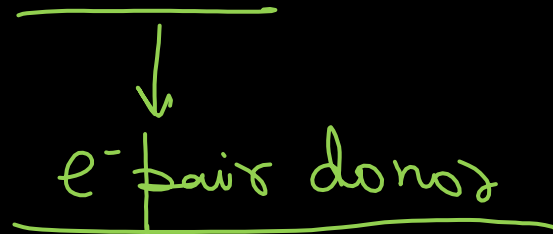
EWG - I | - M
↓
Stabilise

- (😐)

↓
ERG
+ I | + M → destabilise

Characteristics :

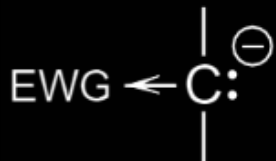
- The C atom is in sp^3 hybridized state with a pyramidal geometry
- The anionic C has 8 electron's in the valence shell with three bond pair and one lone pair
- Since electron rich, behave as a nucleophile.
- Being nucleophile donate electron pair, thus acts like a Lewis base.



Stability of Carbanion

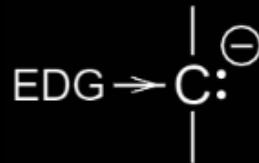
[Electronic effect's contributing towards stability of carbanion]

□ Inductive Effect:



Stabilizes the carbanion

-I

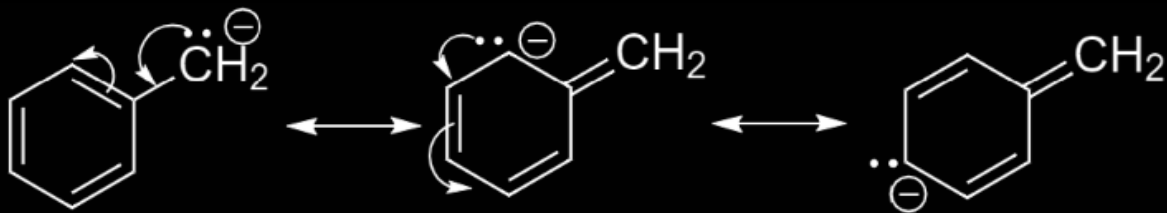


Destabilizes the carbanion

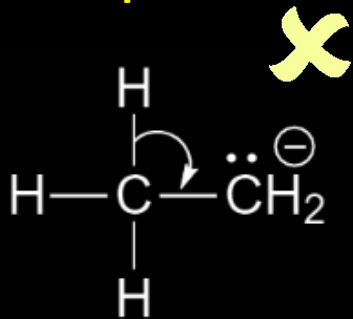
+I

Stability of Carbanion

Resonance :



For example :

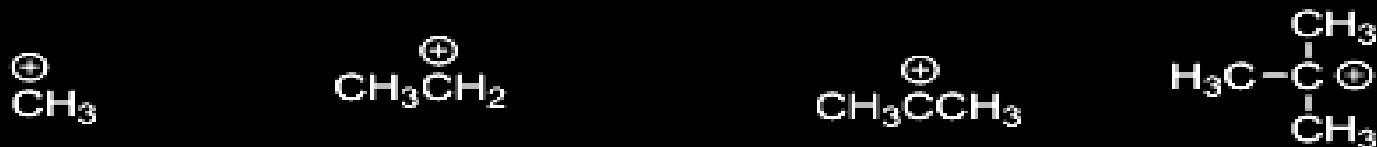


No vacant orbital
available

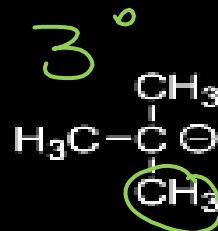
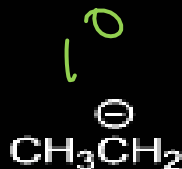
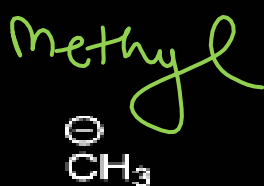
Note: Hyperconjugation does not stabilize the carbanion as there is no vacant orbital available to accommodate delocalized σ e $^-$ pair

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General order of Stability of Carbocation and Carbanion



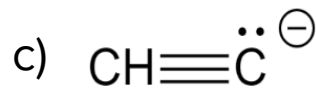
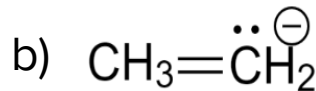
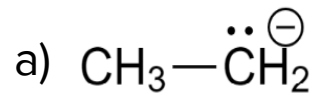
Increasing stability of carbocations



Increasing stability of carbanions

see
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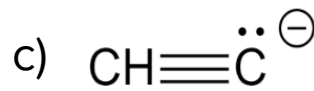
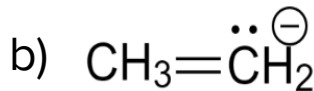
Write the order of stability of following carbanion



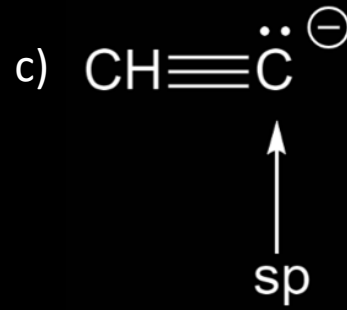
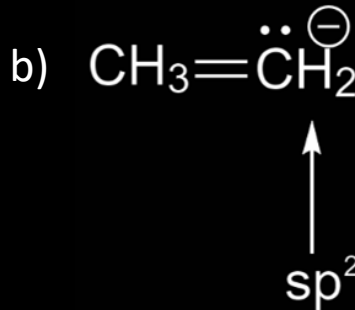
↓
small size
more electronegative

Write the order of stability of following carbanion

a) $\text{CH}_3-\ddot{\text{C}}\text{H}_2^-$ b) $\text{CH}_3=\ddot{\text{C}}\text{H}_2^-$ c) $\text{CH}\equiv\ddot{\text{C}}^-$



Solution :



The order is $(c) > (b) > (a)$

The arrangement of $(\text{CH}_3)_3\text{C}-$, $(\text{CH}_3)_2\text{CH}-$, CH_3CH_2- , when attached to a benzene or an unsaturated group in increasing order electron donating tendency, is

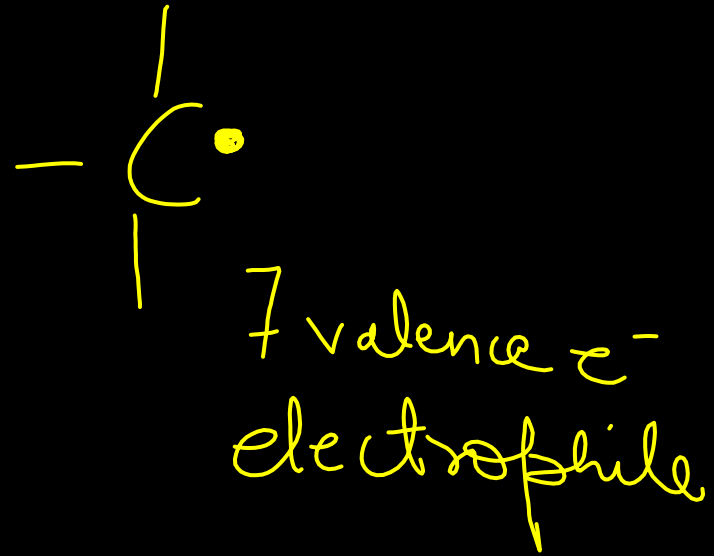
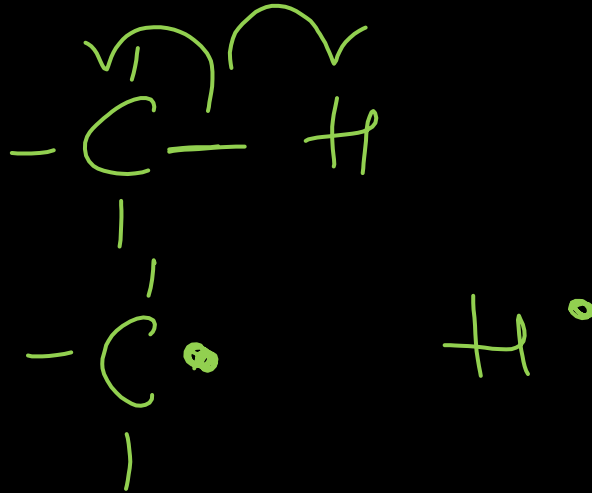
- A $(\text{CH}_3)_3\text{C}- < (\text{CH}_3)_2\text{CH}- < \text{CH}_3\text{CH}_2-$
- B $\text{CH}_3\text{CH}_2- < (\text{CH}_3)_2\text{CH}- < (\text{CH}_3)_3\text{C}-$
- C $(\text{CH}_3)_2\text{CH}- < (\text{CH}_3)_3\text{C}- < \text{CH}_3\text{CH}_2-$
- D $(\text{CH}_3)_3\text{C}- < \text{CH}_3\text{CH}_2- < (\text{CH}_3)_2\text{CH}-$

The arrangement of $(\text{CH}_3)_3\text{C}-$, $(\text{CH}_3)_2\text{CH}-$, CH_3CH_2- , when attached to a benzene or an unsaturated group in increasing order electron donating tendency, is

- A** $(\text{CH}_3)_3\text{C}- < (\text{CH}_3)_2\overset{\curvearrowright}{\text{CH}}- < \text{CH}_3\overset{\curvearrowright}{\text{CH}_2}-$
- B** $\text{CH}_3\text{CH}_2- < (\text{CH}_3)_2\text{CH}- < (\text{CH}_3)_3\text{C}-$
- C** $(\text{CH}_3)_2\text{CH}- < (\text{CH}_3)_3\text{C}- < \text{CH}_3\text{CH}_2-$
- D** $(\text{CH}_3)_3\text{C}- < \text{CH}_3\text{CH}_2- < (\text{CH}_3)_2\text{CH}-$

Free radical

It is the short lived reactive intermediate, neutral carbon specie carrying odd e^- in its valence shell.

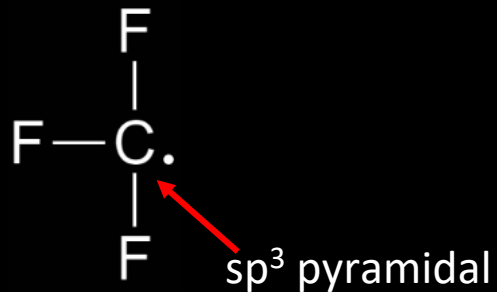


Free radical

Representation, hybridization & geometry :

Reactive Intermediate	$\begin{array}{c} \\ -\text{C}^+ \\ \end{array}$	$\begin{array}{c} \\ -\text{C}\cdot \\ \end{array}$	$\begin{array}{c} \\ -\text{C}^\ominus \\ \end{array}$
Valance e ⁻	6	7	8
Hybridization	sp^2	Between sp^2 & sp^3	sp^3
Shape	Triagonal planar	Slightly pyramidal	Pyramidal

Typical example



Free radical	Hybridization	Location of Odd e^-
$\cdot\text{CH}_3$	sp^2	Pure p
$\cdot\text{CF}_3$	sp^3	sp^3

Typical example

Free radical	Hybridization	Location of Odd e ⁻
Radicals	.CH ₃	.CF ₃
hybridization	sp ²	sp ³
Location of Odd e ⁻	p	sp ³

Typical example

ERG on carbon $\rightarrow sp^2$

EWG on carbon $\rightarrow sp^3$

Resonance stabilised FRs are sp^2 hybridised

Stability: ERG $\uparrow$ stability of C° $(+I/+M)$
EWG $\downarrow$ stability of C° $(-I/-M)$

Resonance Stabilized 3° 2° 1° $\cdot CH_3$ $>$ Vinyl Phenyl

$+I \uparrow$ stability of F ; $HC \uparrow$ stability of FR



Stability of Free Radical

❖ Inductive effect :



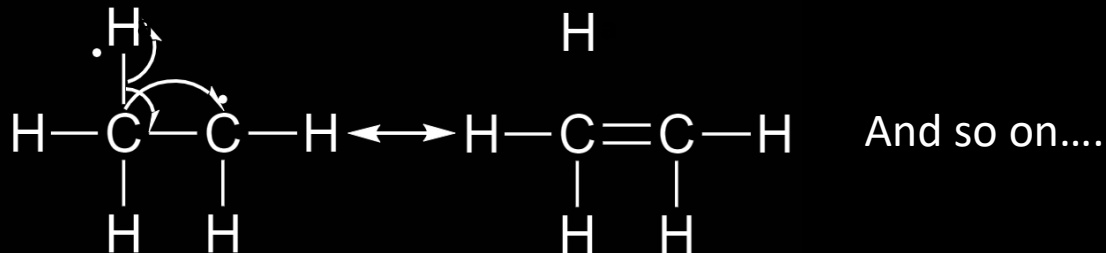
Stabilizes

Destabilizes

EDG increases the stability of free radical.

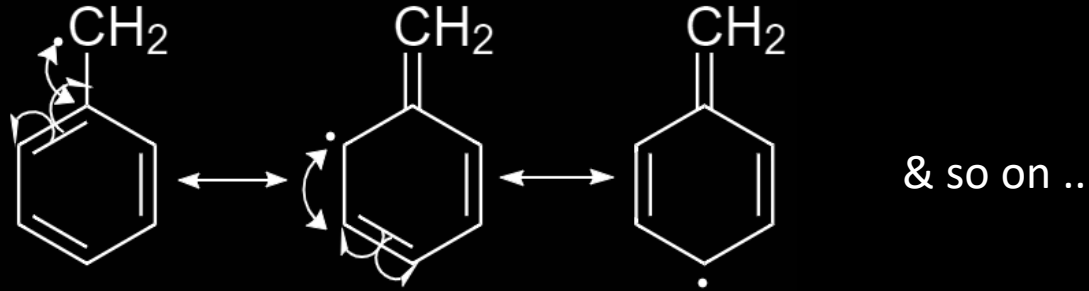
EWG decreases the stability of free radical.

❖ Hyperconjugation effect :



Stability of Free Radical

❖ Resonance effect :



Stability of Free Radical

Conditions that favors the formation of carbon free radical :

H – High temperature

E – Electricity

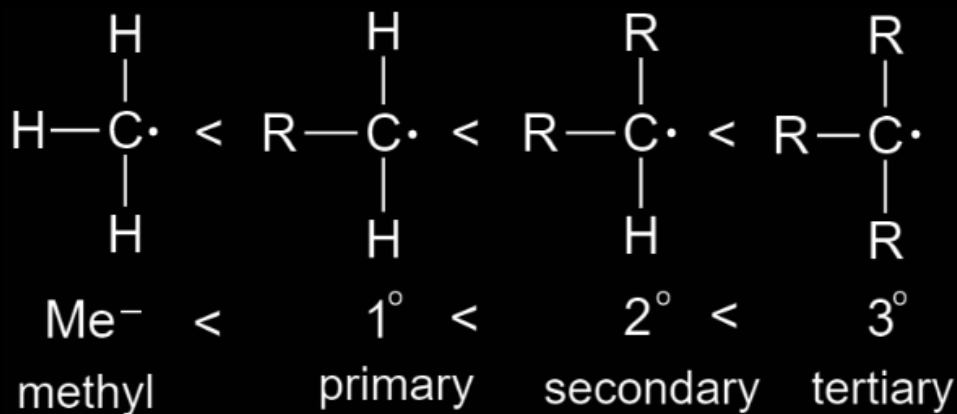
L – Light

P – Peroxide

R – Radical And Non polar solvent

General Stability Order of Free Radical

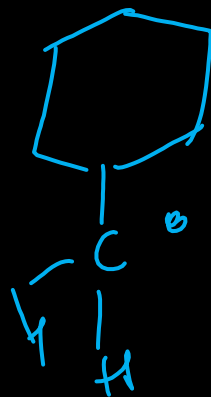
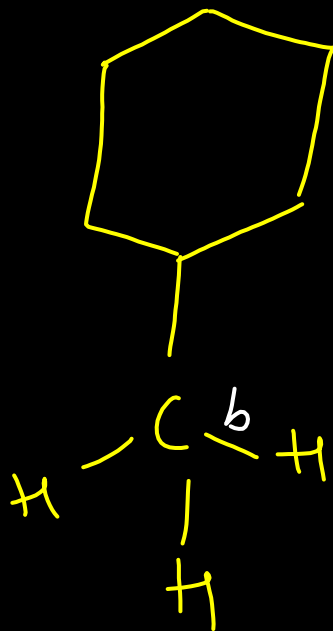
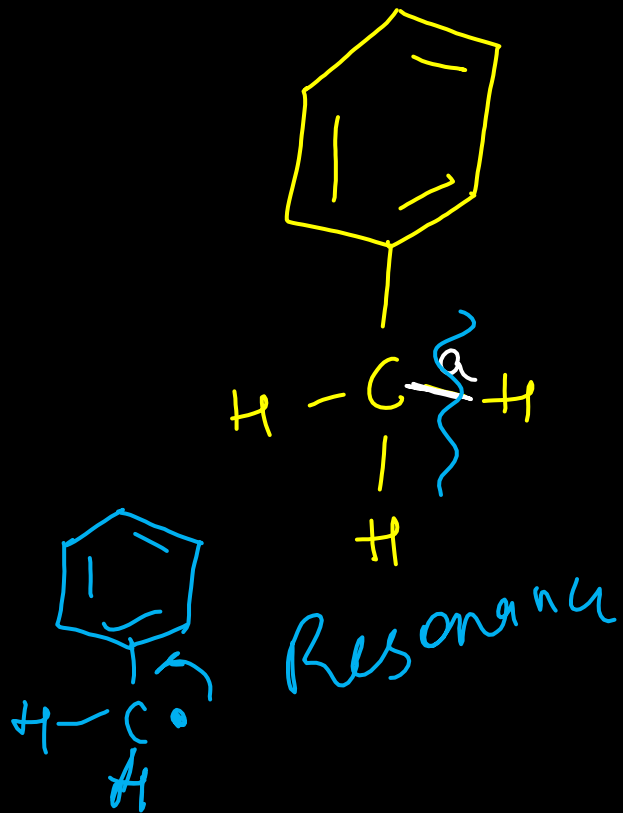
Stability: The order of stability of free radical is the same as that of carbocations i.e., $3^\circ > 2^\circ > 1^\circ$. This order of stability can be easily explained already on the basis of hyperconjugation.



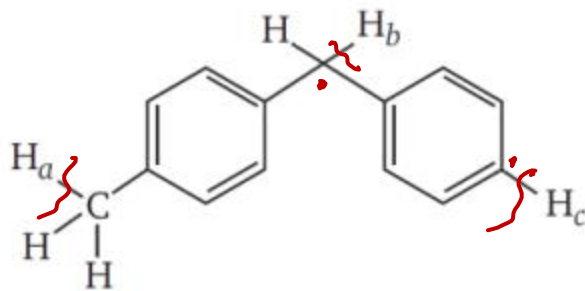
Free radicals are more stable if they are highly substituted.

$$b > a$$

Bond dissociation
Energy



23. For the compound given below, the correct increasing order of C-H bond strength is



(a) $C-H_a > C-H_b > C-H_c$

(b) $C-H_a > C-H_c > C-H_b$

(c) $C-H_b > C-H_a > C-H_c$

(d) $C-H_c > C-H_a > C-H_b$

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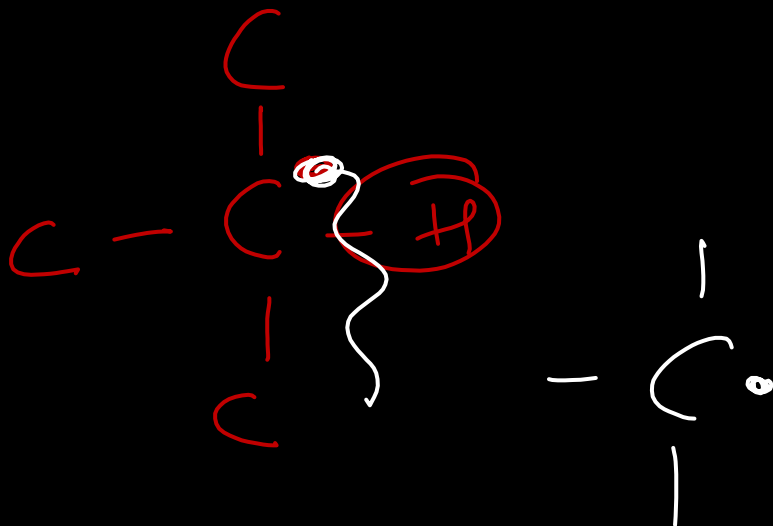
29. Which of the following C-H bonds has lowest bond dissociation energy?

☒ (a) 3° C-H bond

(b) 2° C-H bond

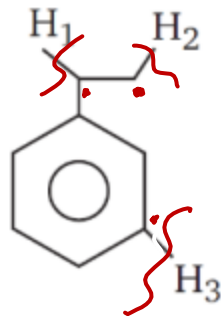
(c) 1° C-H bond

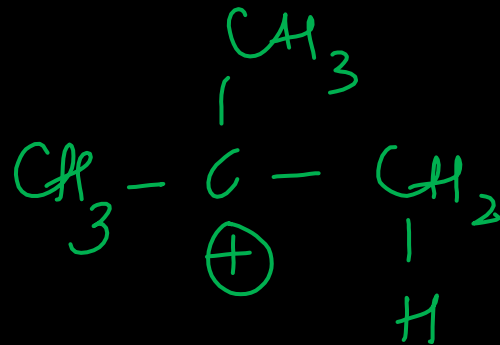
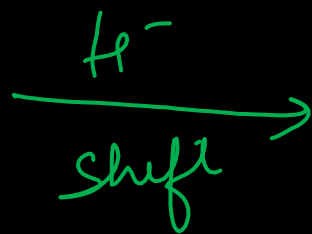
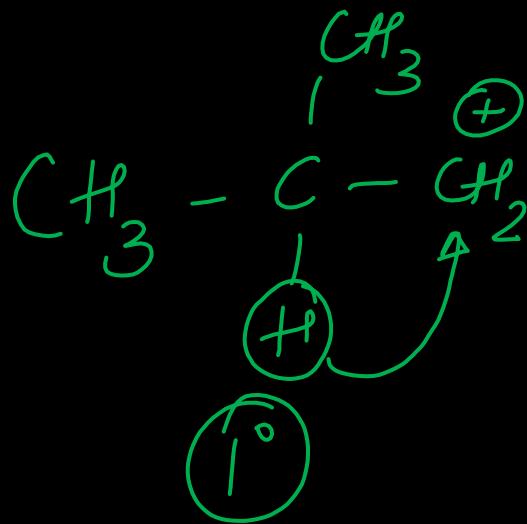
(d) All have same bond dissociation energy



25. The correct order of the bond energy of different C–H bonds in the given compound is

- (a) $\text{C-H}_1 > \text{C-H}_2 > \text{C-H}_3$
- (b) $\text{C-H}_2 > \text{C-H}_1 > \text{C-H}_3$
- ~~(c)~~ $\text{C-H}_3 > \text{C-H}_2 > \text{C-H}_1$
- (d) $\text{C-H}_3 > \text{C-H}_1 > \text{C-H}_2$

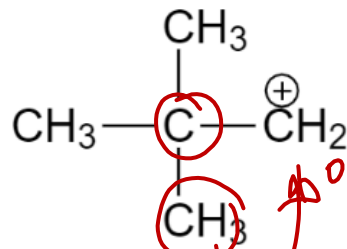




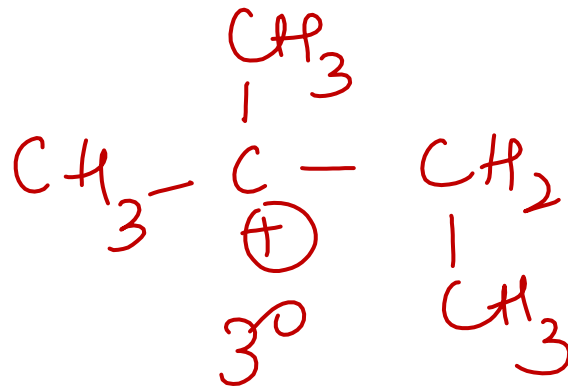
3°

↓
more
stable

The carbocation



stabilizes itself by



A

H^- shift

B

H^+ shift

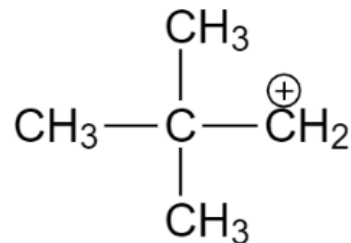
C

CH_3^- shift

D

CH_3^+ shift

The carbocation



stabilizes itself by

A

H⁻ shift

B

H⁺ shift

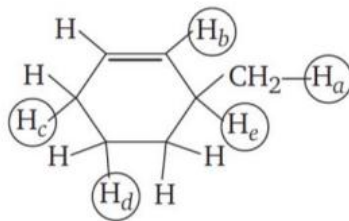
C

CH₃⁻ shift

D

CH₃⁺ shift

31. The correct order of abstraction of hydrogen from the below given compound when treated with halogen is



(a) $e > b > c > a > d$

(b) $b > e = c > a > d$

(c) $e > c > d > a > b$

(d) $b > c > e > d > a$

H.W

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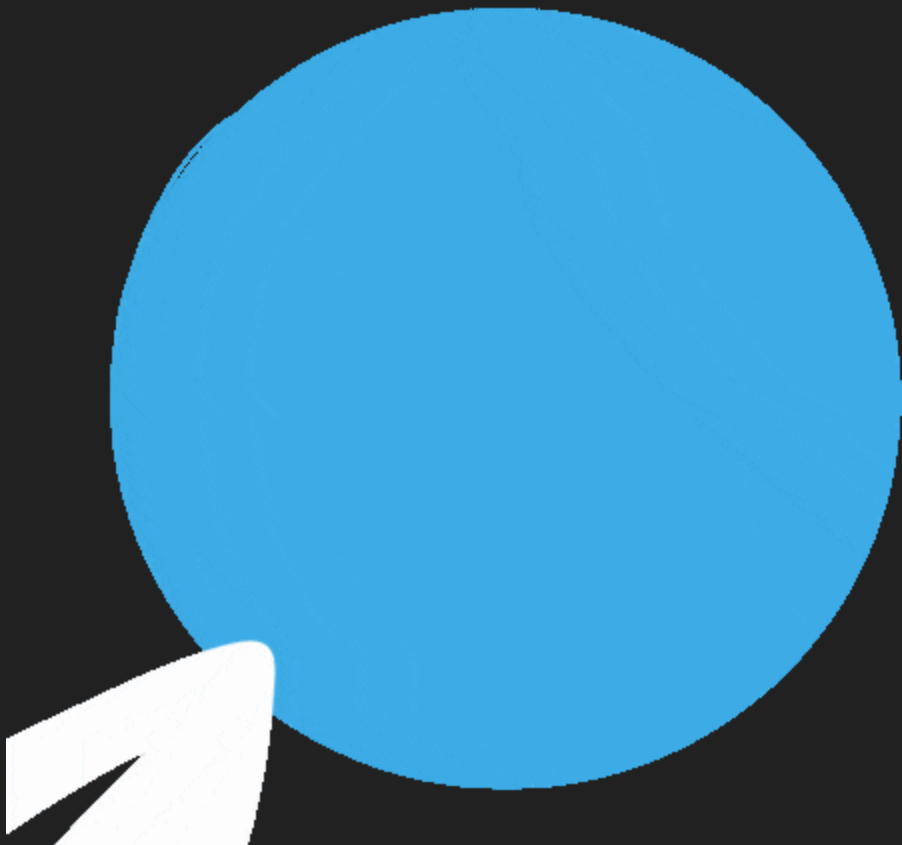
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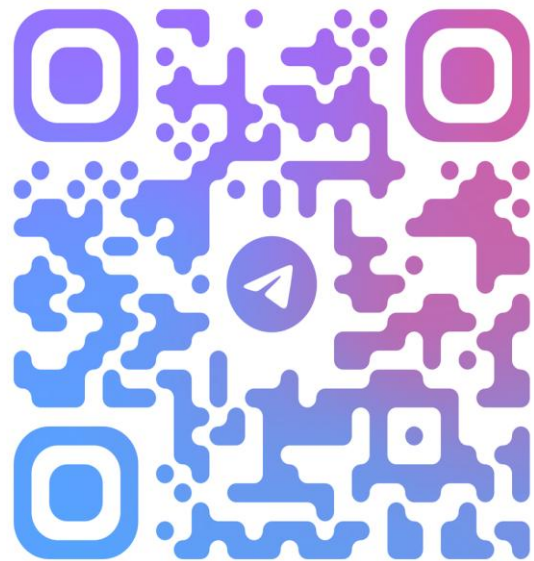
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